

Pressure in Solids and Liquids — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: pressure = force $\div$ area, and in a liquid pressure = density $\times$ g $\times$ depth.
- Take $g = 10 \text{ N/kg}$, and remember that $1 \text{ m}^2 = 10\,000 \text{ cm}^2$.

SECTION 1 – THE EQUATION

- 1.** Write down the equation linking pressure, force and area, and state the unit of pressure. [2]

- 2.** Calculate the pressure in each case.

- (a)** A force of 200 N acting on an area of 4 m^2 . [1]

- (b)** A force of 60 N acting on an area of 0.2 m^2 . [1]

- (c)** A force of 750 N acting on an area of 0.05 m^2 . [2]

- 3.** Calculate the force.

- (a)** A pressure of 500 Pa acting on an area of 6 m^2 . [1]

(b) A pressure of 20 000 Pa acting on an area of 0.02 m². [1]

4. Calculate the area.

(a) A force of 600 N producing a pressure of 300 Pa. [1]

(b) A force of 1500 N producing a pressure of 30 000 Pa. [1]

SECTION 2 – SQUARE CENTIMETRES

5. Convert each area into square metres.

(a) 20 cm² [1]

(b) 400 cm² [1]

(c) 2.5 cm² [1]

6. A force of 60 N acts on an area of 20 cm². Calculate the pressure, in pascals. [3]

7. A box of weight 480 N rests on the floor. Its base measures 0.4 m by 0.5 m. [3]
Calculate the pressure it exerts on the floor.

- 8.** Explain why a tractor is fitted with very wide tyres. [2]

- 9.** A drawing pin has a sharp point at one end and a wide flat disc at the other. [3]
Explain why each end is that shape, referring to force and area.

- 10.** Calculate the pressure due to the water at a depth of 8 m in a lake. The density of [3]
the water is 1000 kg/m^3 .

- 11.** State two quantities that the pressure at a point in a liquid depends on, and one [3]
quantity that it does not depend on.

End of paper. Check every answer has a unit or a form the question actually asked for.